[7.1.

Let $\{f_n : X \to Y\}$ be a uniformly convergent sequence of bounded functions.
Prove it is uniformly bounded.

Proof. Given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow \sup_X |f_n - f| \leq \varepsilon$, where f is the limit function of f_n.
Therefore, $\sup_X |f| \leq \sup_X |f_N| + \varepsilon$, the limit function f is bounded. Now,
for each $1 \leq i < N$, let M_i be a bound of f_i. Then, for all $k \in \mathbb{N}$,

$$\sup |f_k| \leq \operatorname{Max}\{M_1, \dots, M_{N-1}, \sup |f| + \varepsilon\}.$$

[7.2-3. Let $f_n, g_n : X \to Y$ be functions.

1). If $\{f_n\}, \{g_n\}$ converges uniformly, then $\{f_n + g_n\}$ converges uniformly.
2). If $\{f_n\}\ \{g_n\}$ converges uniformly of bounded functions, then $\{f_n g_n\}$ converges uniformly.
3). Find counter example for 2), the boundedness omitted.

Proof. 1) can be shown directly, by $\sup |f_n + g_n| \leq \sup |f_n| + \sup |g_n|$.
But 2) is needed the boundedness condition: for example,

$$f_n : (0,1] \to \mathbb{R} : x \mapsto \frac{n}{n+1} \cdot \frac{1}{x} \qquad \left(\sup \left|f_n - \frac{1}{x}\right| = \frac{1}{n+1} \to 0\right)$$

$$g_n : (0,1] \to \mathbb{R} : x \mapsto \frac{n+1}{n^2} \cdot \frac{1}{x} \qquad \left(\sup \left|g_n - \frac{1}{x}\right| = \frac{1}{n} + \frac{1}{n^2} \to 0\right)$$

But $f_n g_n(x) = \frac{1}{nx}$ is Not uniformly convergent, because $\sup |f_n g_n - 0| = \infty$, but $f_n g_n \to 0$.

Now, suppose that f_n and g_n are bounded (of course, uni. convergent.)
By the above problem, f_n and g_n are uniformly bounded. Now,

$$|f_n g_n - fg| = |(f_n - f)\cdot g_n + (g_n - g)\cdot f| \leq |g_n||f_n - f| + |f||g_n - g|$$

gives the result. (More specifically, given $\varepsilon > 0$, there is $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow \sup |f_n - f|, \ \sup |g_n - g| < \varepsilon.$$

Therefore, for these $n \geq N$,

$$\sup |f_n g_n - fg| \leq 2M\varepsilon, \text{ where } M \text{ is bound of } |g_n| \text{ and } |f|.$$

7.4. Consider a map defined on Real Numbers

$$f(x) = \sum_{n=1}^{\infty} \frac{1}{1+n^2x}$$

1). Find $x \in \mathbb{R}$ for the series Converges absolutely

2) Find interval for the series Converges Uniformly

3) Find interval for the Converges Uniformly fails.

4). If the series Converges, is f Continuous? 그만 물어봐

5) 〃 , f bounded?

Sol). Since $\left|\frac{1}{1+n^2x}\right| \le \frac{1}{n^2x}$ if $x>0$, for all $x>0$, it Converges absolutely.

7. 20. If f is continuous on $[0,1]$ and

$$\int_0^1 f(x)\cdot x^n \, dx = 0 \quad \text{for all } n = 0, 1, \cdots$$

Then $f = 0$.

sol). By the Weierstrass Thm, there exists a sequence of polynomial $\{P_n\}$ s.t $P_n \rightrightarrows f$.
And, since f and P_n are continuous on the compact set $[0,1]$, they are bounded.
Therefore $f \cdot P_n \rightrightarrows f \cdot f = f^2$. Now, clearly $f \cdot P_n \in \mathcal{R}$, thus

$$\lim_{n\to\infty} \int_0^1 f(x) \cdot P_n(x) \, dx = \int_0^1 \lim_{n\to\infty} f(x) P_n(x) \, dx = \int_0^1 [f(x)]^2 \, dx$$

But by the assumption, for each $n \in \mathbb{N}$, $\int_0^1 f(x) P_n(x) \, dx = 0$.

Therefore, $\int_0^1 [f(x)]^2 \, dx = 0$, thus $f^2 = 0$, by $f^2 \geq 0$. Consequently $f = 0$.